What is photosynthesis response?

The rate at which organisms can take in energy is limited, either by external or internal constraints. The relationship between photon flux density and plant photosynthetic rate is called photosynthetic response. Herbs and short-lived perennial shrubs from sunny habitats have high maximum photosynthetic rates that level on at high light intensities. The lowest maximum rates of photosynthesis occur among plants from shady environments.

A plant's photosynthetic response is the rate at which the plant utilizes sunlight to convert water and carbon dioxide into glucose. The glucose is then used as food energy for the plant to fuel its growth, bud formation, flowering, and seed production.

What are stop codon

Stop codons are also called nonsense codons because they do not code for an amino acid and give signal for end of protein synthesis. e.g., UAG, UAA, UGA.

How predators and parasites play role in population control?

Predators and parasites have been used to control populations of insects that attack crops or to control invasive weeds.

Recent research in Kenya has shown that a crayfish, Procambarus clarkii, controls the snails that act as intermediate hosts for Schistosoma, a highly pathogenic human parasite.

Preliminary results indicate that crayfish successfully control host snails in the artificial impoundments used for livestock watering and domestic water, important sources of infection by Schistosoma.

How nutrients enrichment disturb the aquatic and terrestrial ecosystem?

Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems. Nitrate concentration and export by the earth's major rivers correlate directly with human population density. Human disturbance also increases export of phosphorus from aquatic catchments. Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems.

Phylogenetic tree length indicate

In some trees, the length of a branch can reflect the number of genetic changes that have taken place in a particular DNA sequence in that lineage. In other trees, branch length can represent chronological time, and branching points can be determined from the fossil record.

Transcription & translation

Transcription is the synthesis of RNA from a DNA template where the code in the DNA is converted into a complementary RNA code. **Translation** is the synthesis of a protein from an mRNA template where the code in the mRNA is converted into an amino acid sequence in a protein.

Role of bacteria

The ecosystem, both on land and in the water, depends heavily upon the activity of bacteria. The cycling of nutrients such as carbon, nitrogen, and sulfur is completed by their ceaseless labor. Decomposition is the breakdown of these organisms, and the release of nutrients back into the environment, and is one of the most important roles of the bacteria.

Characters of animals likes protists

Protists are a strange and diverse group of organisms. They are unicellular eukaryotes. They are not animals, plants, fungi, or bacteria. As a result, taxonomists, scientists that name, describe, and classify organisms, created a special classification to group these misfits: Kingdom Protista. The protists include organisms that are plant-like (such as algae), organisms that are animal-like (such as amoebas), and organisms that are fungus-like (such as slime molds). It turns out that protists from each of these groups are important to life on Earth.

Food web with its structure in community

A food web summarizes the feeding relations in a community.

- The earliest work on food webs concentrated on simplified communities in areas such as the Arctic islands. However, researchers such as Charles Elton (1927) soon found that even these so-called simple communities included very complex feeding relations.
- The level of food web complexity increased substantially, however, as researchers began to study complex communities. Studies of the food webs of tropical freshwater fish communities

revealed highly complex networks of trophic interaction that persisted even in the face of various simplifications.

- A focus on strong interactions can simplify food web structure and identify those interactions responsible for most of the energy flow in communities.

General character of diatoms

Diatoms

➢ Float in Water

➢ Photosynthesis

➢ Unicellular

➢ Silica Shells

➢ Commercial value: Insulating materials, Abrasives, Ceramics, Filtering

Cladistics and clade

Cladistics is a method of taxonomic classification based on their evolutionary history. It was developed by Willi Hennig, a German entomologist, in 1950.

A clade is a group of species that includes an ancestral species and all its descendants.

Differentiate among Ecology, industrial ecology and Ecology Engineering

ECOLOGY:

Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).

INDUSTRIAL ECOLOGY:

- The design of the industrial infrastructure such that it consists of a series of interlocking "technological ecosystems" interfacing with global natural ecosystems.
- Industrial ecology takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in model from conquering nature to becoming nature.

ECOLOGICAL ENGINEERING:

- Unlike industrial ecology, the focus of Ecological Engineering is on the manipulation of natural ecosystems by humans for our purposes, using small amounts of supplemental energy to control systems in which the main energy drives are still coming from non-human sources.
- It is the design of new ecosystems for human purposes, using the self-organizing principles of natural ecosystems.

Exotic predators' effects on structure of food web

Exotic predators can collapse and simplify the structure of food webs. Introduced fishes have devastated the native fishes of Lake Atitlan and Gatun Lake in Central America.

Examples

The influence of the Nile perch on the fish community of Lake Victoria is enmeshed with massive changes in the lake's ecosystem. Introduction of the Nile perch is rapidly reducing the species-rich fish fauna of Lake Victoria to a community dominated by a handful of species.

Population Growth with example

Population growth is the increase in the number of individuals in a population. Global human population growth amounts to around 83 million annually, or 1.1% per year.

In the presence of abundant resources, populations can grow at geometric or exponential rates.

Lyell's mistake while theorizing the natural selection

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples…more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

Why the study of genetic in ecosystem management is necessary?

• Loss of genetic diversity can imply lack of evolution and premature extinction.

• Fitness decreases with reduced genetic variation.

• Populations of endangered species are small and tend to lose genetic diversity.

• When genetic variation disappears the basis for life on earth becomes impoverished

What's happen to the organism have limited access nutrients?

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

Factors Influence decomposition of environment.

Decomposition rate is influenced by temperature, moisture, and chemical composition of litter and the environment. The rate of decomposition affects the rate at which nutrients, such as nitrogen and phosphorus, are made available to primary producers. Rates of decomposition in terrestrial ecosystems are higher under warm, moist conditions. The rate of decomposition in terrestrial ecosystems increases with nitrogen content and decreases with the lignin content of litter. The chemical composition of litter and the availability of nutrients in the surrounding environment also influence rates of decomposition in aquatic ecosystems.

Explain optimal foraging theory.

Optimal foraging theory attempts to model how organisms feed as an optimizing process.

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

Who worked on developing paleontology?

Smith, Buckland and Cuvier himself (along with Brongniart) were developing paleontology

Examples of foundation species

A foundation species plays a unique, essential role in creating and defining a community. Often, foundation species act by modifying the environment so that it can support the other organisms that form the community, Foundation species are sometimes referred to as 'Ecosystem Engineers'. Examples of Foundation Species are; Corals, Earthworms, Beavers and many more

Hibernation with examples

Animals can go into a state of reduced metabolism that may last several months. If this state occurs mainly in winter, it is called hibernation

Examples: bats, snake and bears.

Can humans act as keystone species give example?

Humans have acted as keystone species in communities. People have long manipulated food webs both as a consequence of their own feeding activities and by introducing or deleting species from existing food webs. In addition, many of these manipulations have been focused on keystone species.

General trait of green algae.3mrk

❖ Most freshwater

❖ Food Source – called Plankton

❖ Some are: Free-living, Colony, Filament

Share derived, and ancestor derived.5mrk

- In comparison with its ancestor, an organism has both shared and different characteristics
- A shared ancestral character is a character that originated in an ancestor of the taxon
- A shared derived character is an evolutionary novelty unique to a particular clade
- A character can be both ancestral and derived, depending on the context; it is useful to know in which clade a shared derived character first appeared
- An out group is a species or group of species that is closely related to the in group , the various species being studied
- Systematics compare each in group species with the out group to differentiate between shared derived and shared ancestral characteristics.

Advantage of multicell.5mrk

We know from the fossil record that multi-celled organisms were present during the Proterozoic, but we do not know exactly when they appeared What seem to be some kind of multi-celled algae appear in the 2.1-billionyear-old fossils from the Negaunee Iron Formation in Michigan as carbonaceous filaments from 1.8 billionyear-old rocks in China as somewhat younger carbonaceous impressions of filaments and spherical forms

For something on the order of 1.5 billion years all organisms were single-celled and life seems to have thrived. In fact, single-celled organisms are quite good at what they do but what they do is very limited. Single celled organisms cannot grow very large, because as size increases proportionately less of a cell is exposed to the external environment in relation to its volume and the proportion of surface area decreases. Transferring materials from the exterior to the interior

becomes less efficient. Multi-celled organisms live longer, since cells can be replaced and more offspring can be produced. Cells have increased functional efficiency when they are specialized into organs with specific capabilities.

Carnivore and detritivores 3mrk.

Carnivores are animals that prey on other animals in order to survive**.** The meat is the most essential part of their diet.

A detritivore is a heterotrophic organism, which obtains its nutrition by feeding on detritus. Detritus is the organic matter made up of dead plant and animal material.

Is replication involve in the transformation of RNA form DNA?

Nutrient cycle

A nutrient cycle refers to the movement and exchange of organic and inorganic matter back into the production of living matter. The process is regulated by the food web pathways, which decompose organic matter into inorganic nutrients. Nutrient cycles occur within ecosystems.

Sigmoidal population

As population size increases, population growth eventually slows and then ceases, producing a sigmoidal, or S-shaped, population growth curve. Sigmoidal population growth can be modelled by the logistic growth equation, a modification of the exponential growth equation that includes a term for environmental resistance.

Reproduction in leeches

• Leeches are simultaneous hermaphrodites that lack a free-living larvae stage

• Fertilization is internal through copulation

• Development occurs in a cocoon similar to the Oligochaetes

Maximum Parsimony and Maximum Likelihood

• Maximum parsimony assumes that the tree that requires the fewest evolutionary events (appearances of shared derived characters) is the most likely

• The principle of maximum likelihood states that, given certain rules about how DNA changes over time, a tree can be found that reflects the most likely sequence of evolutionary events

Lamarck contribution

Lamarck is best known for his contributions to evolution, or Lamarckism, which suggests organisms acquire or lose traits based on how much they use them in their lives.

Exotic predators and structure of food web

Exotic predators can collapse and simplify the structure of food webs. Introducing new fish wiped out the local fish population in Central America lakes. Examples the effect of the Nile perch on the fish community of Lake Victoria is destroy the ecosystem of lakes.

Nutrient Loss from Ecosystem?

Major pathways in which these nutrients are lost include: soil erosion, leaching and gaseous losses. Another loss of nutrients from ecosystems is due to denitrification.